Import Data using Transform Maps

Spreadsheet Employee Data Import in ServiceNow

Final Project Report

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Prepared from the supplied Import Data using Transform Maps documentation. The report uses the project screenshots from that document. Individual team roles were not supplied.

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Project Report Format

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1. INTRODUCTION

1.1 Project Overview

This micro project imports structured employee data from an external Excel spreadsheet into ServiceNow. It uses an Import Set as a staging area and a Transform Map to connect spreadsheet fields to the Employee Test table. Records are checked after transformation to confirm that the source data reaches the correct target fields.

The employee dataset contains Employee ID, Employee Name, Email, Department and Location. Google Sheets is used to prepare the data and export Sample Spreadsheet.xlsx. The target table is Employee Test (u_employee_test), and the staging table is Employee Import (u_employee_import).

1.2 Purpose

The purpose is to demonstrate an end-to-end employee data import process: prepare the file, create the target fields, load the data, map the fields, run the transformation and verify the results. Employee ID coalesce supports repeat imports by updating matching employees and inserting new employees without duplicating existing IDs.

The project also provides Employees by Department, Employees by Location and Employee List Report. These reports are placed on Employee Analytics Dashboards so HR teams and administrators can review employee data from one view.

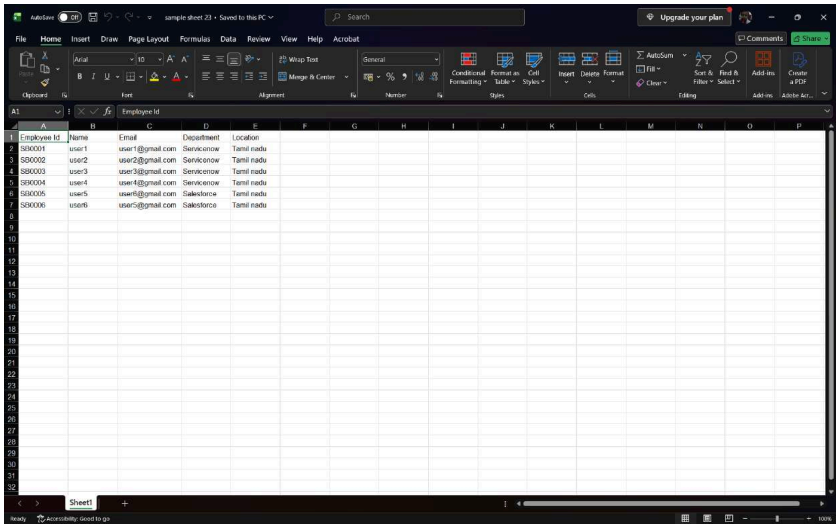


Figure 1. Employee spreadsheet supplied in the source documentation.

2. IDEATION PHASE

2.1 Problem Statement

Organisations receive employee data in spreadsheets and need to move it into ServiceNow accurately. Manual entry increases effort and creates opportunities for incorrect field values. Repeated uploads can produce duplicate records unless an existing employee is identified before insertion.

2.2 Empathy Map Canvas

The following canvas expresses the needs implied by the project workflow. It is a design interpretation of the supplied documentation, not a record of customer interviews.

Canvas area	HR team or administrator perspective
Says	Employee records should be accurate and easy to find.
Thinks	Will a repeated file create duplicates or overwrite the wrong record?
Does	Prepares spreadsheets, uploads files, maps fields, checks records and reviews reports.
Feels	Concern about errors before import; confidence after reviewing successful results.
Pains	Manual work, duplicate records and separate employee summaries.
Gains	Reusable imports, reliable updates and a central employee dashboard.

2.3 Brainstorming

The project combines six practical ideas: spreadsheet preparation, staging through Import Sets, reusable field mapping, Employee ID coalesce, import-history verification, and native ServiceNow reports. The selected workflow uses built-in platform features and follows the source document’s nine phases.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Action	Expected outcome
Prepare	Create five employee columns and export Excel.	A consistent source file is ready.
Configure	Create Employee Test and its String fields.	The target can store employee data.
Import	Load the file into Employee Import.	Rows reach the staging table.
Transform	Map fields and run Sample Spreadsheet Import.	Target rows are created or updated.
Validate	Review history and compare employee values.	Errors or unexpected changes are identified.
Analyse	Create reports and add them to the dashboard.	Employee distributions and details are visible.

3.2 Solution Requirement

Functional requirements: import Excel data; store five employee fields; map source fields to target fields; execute transformations; validate records; identify existing employees by Employee ID; support repeated uploads; create three reports; and restrict dashboard sharing through users, groups and roles.

Non-functional requirements: clear forms and list columns, accurate mapping, duplicate prevention, authorised access, reusable configurations and bulk handling. The source gives no measured response-time target, maximum row capacity or availability benchmark.

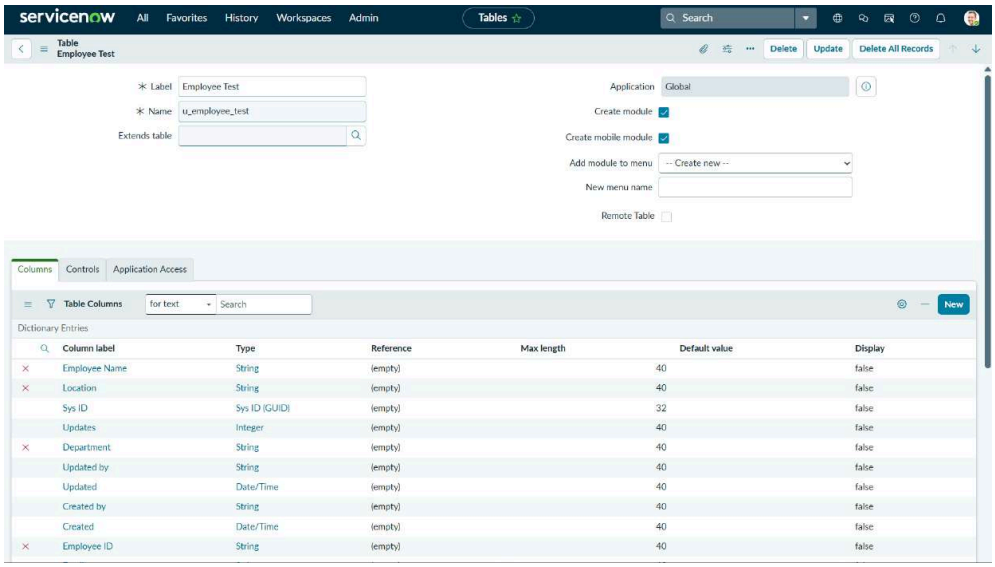


Figure 2. Employee Test form with the project fields.

3. REQUIREMENT ANALYSIS

3.3 Data Flow Diagram

Employee spreadsheet -> Load Data -> Employee Import -> Sample Spreadsheet Import -> Employee Test -> Reports -> Employee Analytics Dashboards. Import history provides a separate validation view after transformation. Employee ID coalesce determines whether an incoming row matches an existing employee.

3.4 Technology Stack

Component	Technology or configuration
Data preparation	Google Sheets and Excel .xlsx export
User interface	ServiceNow web forms and lists
Staging storage	Employee Import / u_employee_import
Target storage	Employee Test / u_employee_test
Import logic	Import Sets, Transform Maps and Mapping Assist
Duplicate handling	Employee ID field map with coalesce enabled
Reporting	Native ServiceNow Pie, Bar and List reports
Dashboard	Employee Analytics Dashboards and sharing settings

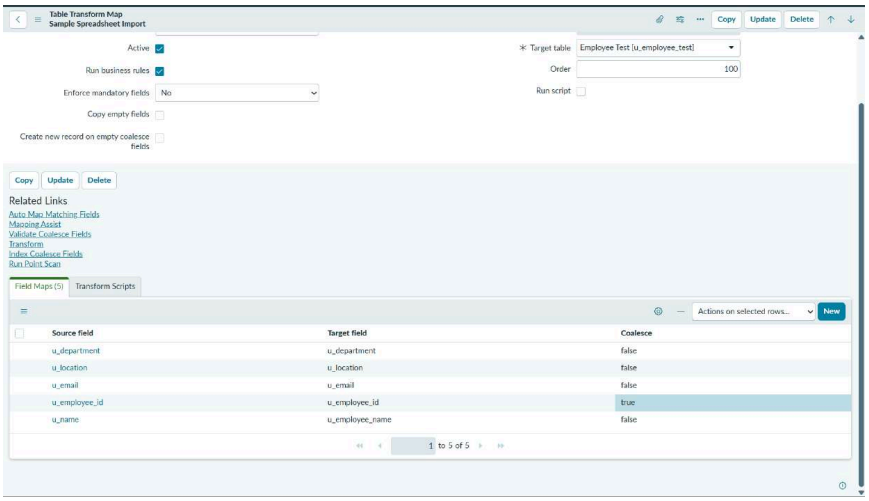


Figure 3. Source-to-target field mappings in the supplied document.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits HR teams and ServiceNow administrators who already receive employee spreadsheets. It uses the same file-based process while replacing individual record entry with bulk loading and repeatable transformations. Coalesce addresses the specific risk of duplicate IDs during repeated imports.

4.2 Proposed Solution

Create Employee Test with Employee ID, Employee Name, Email, Department and Location as String fields. Load the spreadsheet into Employee Import. Configure Sample Spreadsheet Import, verify all five field maps and execute the transformation. Enable coalesce on Employee ID before testing repeated uploads. Create the three employee reports and add them to the shared dashboard.

4.3 Solution Architecture

The local part contains the browser and employee spreadsheet. The ServiceNow instance hosts the staging table, Transform Map, target table, import history, reports and dashboard. No separate React frontend, Node.js backend, MongoDB database, AI model or external API is described in the source.

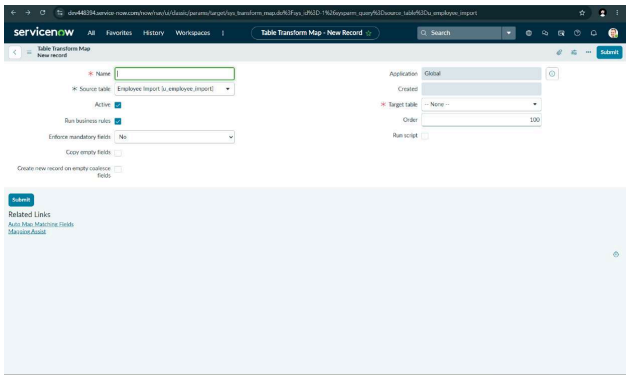


Figure 4. Sample Spreadsheet Import configuration and target table selection.

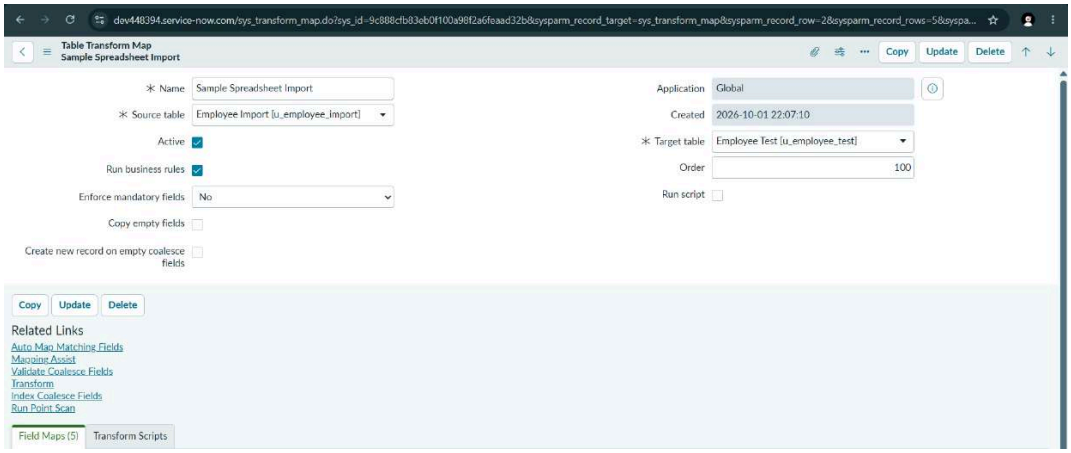


Figure 5. Employee ID field map configured for coalesce.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The plan below groups the source workflow into two illustrative sprints. Story points and six-day sprint durations are planning estimates, not measured delivery results. Team roles, calendar dates and completed-sprint records are not supplied.

Sprint	Tasks	Points
Sprint 1	Spreadsheet preparation (2); target table (3); Load Data (1); Transform Map and field mapping (3); first transformation and validation (3).	12
Sprint 2	Employee ID coalesce (5); changed/new/repeated data tests (5); three reports (5); dashboard and restricted sharing (5).	20

Total planned effort = 32 story points. Planned average capacity = 32 / 2 = 16 story points per sprint. Actual velocity must be calculated from completed story points after completed sprints. The dependency order is target fields and source file, then import and transformation, then repeat-import checks, then reports and dashboard.

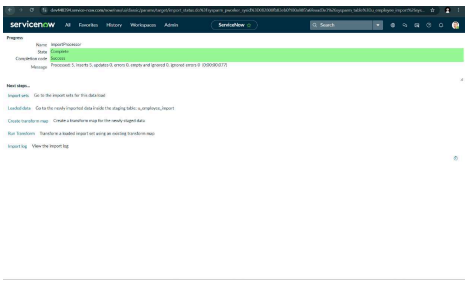


Figure 6. Successful data loading and the Create Transform Map action.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional checks compare spreadsheet values with Employee Test records, inspect source-to-target maps and review import history. The source demonstrates repeat imports using Employee ID coalesce. These examples provide project evidence, but a separately signed UAT report is not supplied.

Check	Source-documented result or testing status
Initial import	The source shows successful transformation and employee records in Employee Test.
Changed and new data	Four incoming rows produce two inserted records and two updated records.
Repeat identical data	The source example reports zero inserts, zero updates and four ignored rows.
Reports	Department Pie, Location Bar and Employee List Report are created.
Dashboard	The three reports are consolidated into Employee Analytics Dashboards.
Performance measures	Import duration, throughput, maximum volume and concurrent load were not measured in the supplied document.

A future performance run should record input row count, file size, transformation duration and error count. Acceptance thresholds should be agreed before testing. Negative tests should include incomplete mappings, inconsistent source values and unauthorised dashboard access.

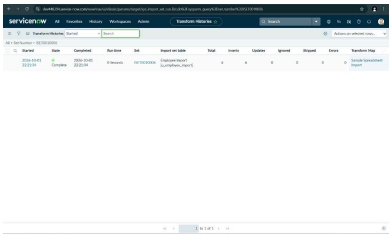


Figure 7. Transform history showing the changed and new employee import.

7. RESULTS

7.1 Output Screenshots

The screenshots in this section come directly from the supplied documentation. They show the project output rather than generated demonstrations.

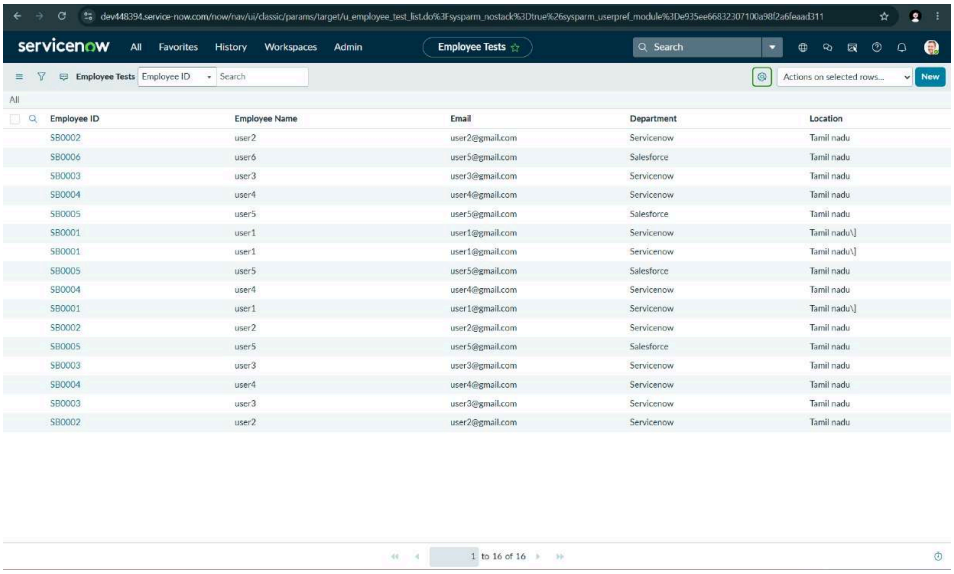


Figure 8. Employee records after the initial import.

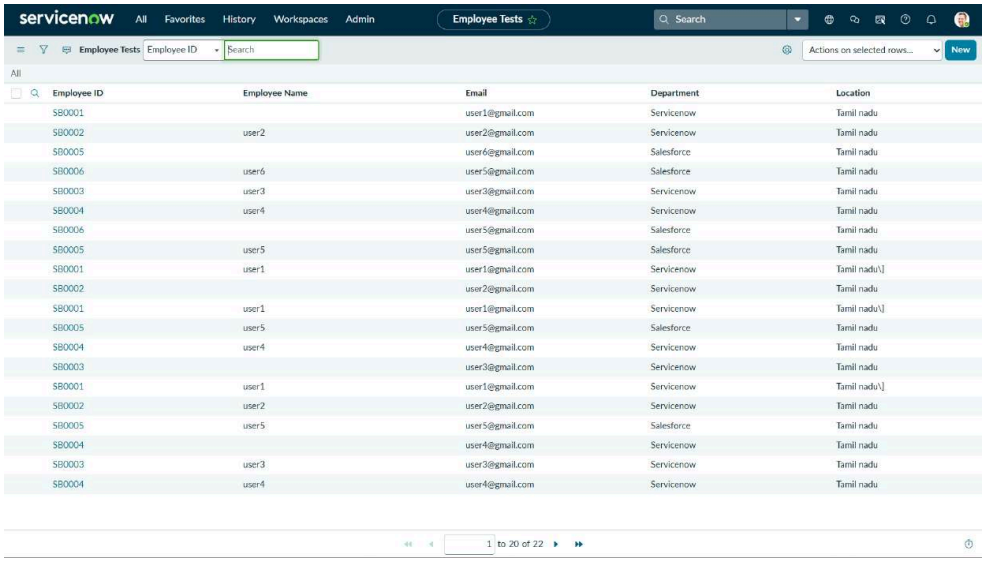


Figure 9. Employee Test records after changed and new rows are transformed.

7. RESULTS

7.1 Output Screenshots

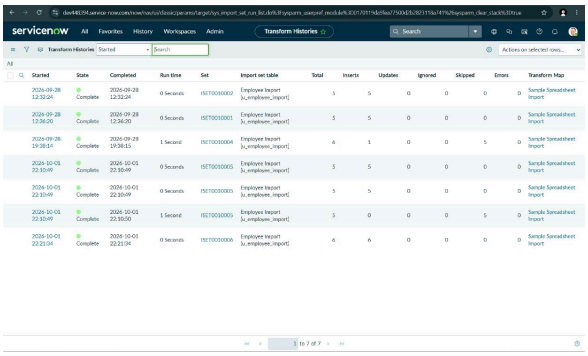


Figure 10. Identical-file reimport history showing the source-reported ignored rows.

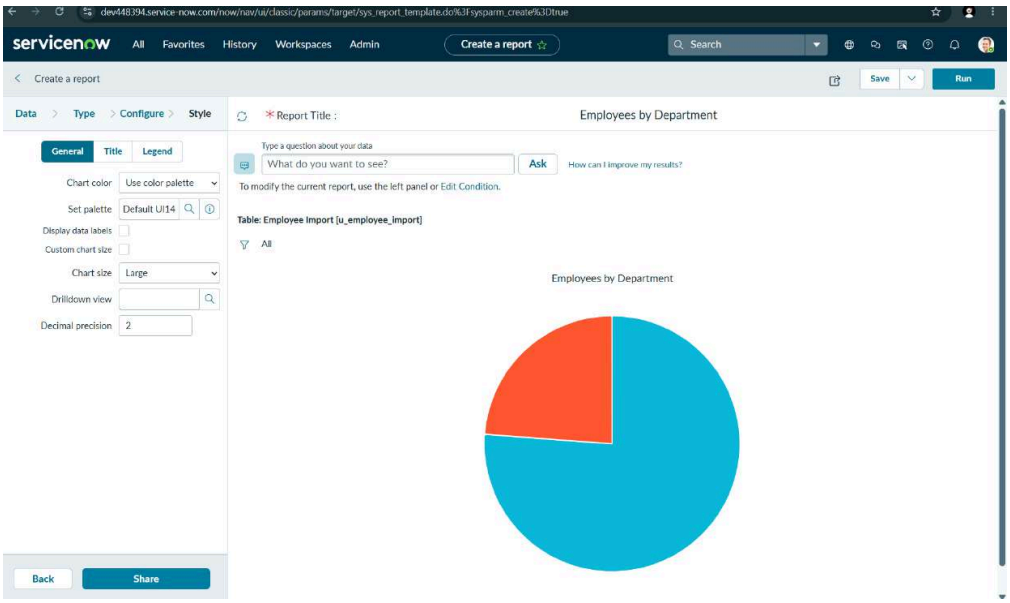


Figure 11. Employees by Department Pie chart.

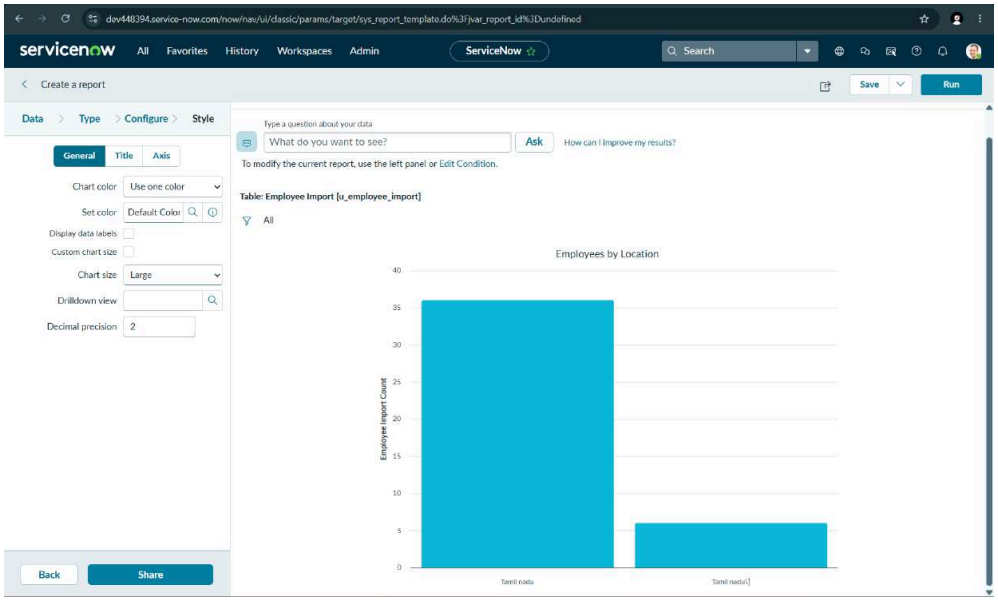


Figure 12. Employees by Location Bar chart.

7. RESULTS

7.1 Output Screenshots

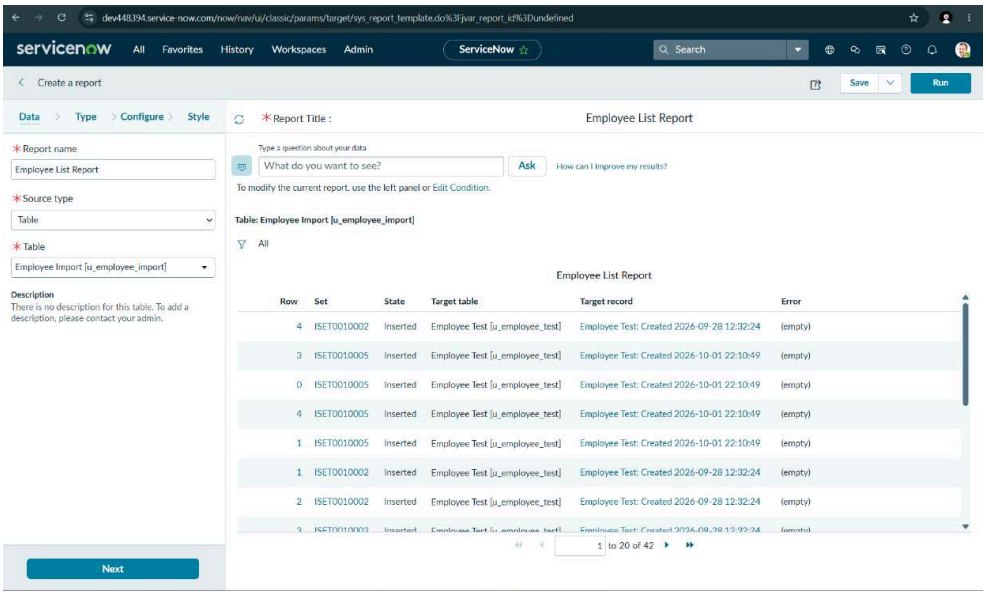


Figure 13. Employee List Report with the five required columns.

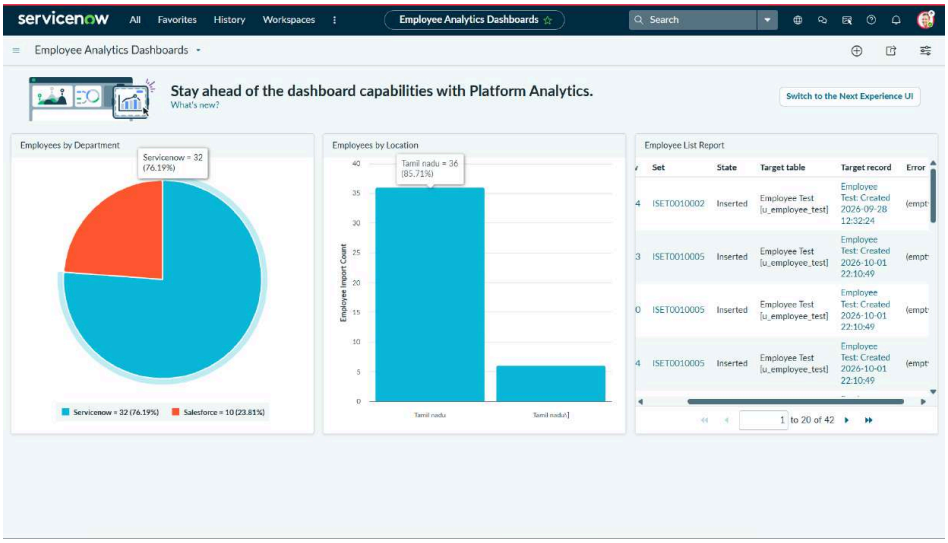


Figure 14. Employee Analytics Dashboards with the three reports.

The result is a reusable employee import workflow with duplicate prevention and native employee reporting. The charts use Count aggregation, while the employee list provides the underlying record details.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages or limitations
Bulk import reduces individual data entry.	Incorrect source columns or maps can produce incorrect target data.
Saved mappings can be reused.	The ServiceNow instance and authorised permissions are required.
Employee ID coalesce prevents duplicate IDs.	Missing or inconsistent employee IDs weaken matching.
History supports record verification.	Formal performance benchmarks and UAT sign-off are not supplied.
Three reports provide a central employee view.	The project does not document automated scheduled imports or external integrations.

9. CONCLUSION

The project demonstrates structured spreadsheet import into ServiceNow using Import Sets and Transform Maps. Employee ID coalesce makes repeat imports reliable by updating existing employees and inserting new employees. Record validation, employee reports and a shared dashboard complete the workflow described in the source document.

10. FUTURE SCOPE

Potential enhancements include validating required IDs and email formats before transformation, scheduling recurring imports, adding clearer import-error summaries, measuring large-file performance, and extending employee reporting. These are proposed additions and are not claimed as implemented features.

11. APPENDIX

Source Code (if any)

The supplied project uses ServiceNow table, Import Set, Transform Map, report and dashboard configurations. No standalone source-code repository or custom import script is provided.

Dataset Link

The dataset is the employee spreadsheet prepared in Google Sheets and exported as Sample Spreadsheet.xlsx. Its five columns are Employee ID, Employee Name, Email, Department and Location. No shareable dataset URL is supplied.

GitHub & Project Demo Link

No GitHub URL, public demo URL or ServiceNow instance URL is provided in the source documentation. Screenshots serve as the supplied project demonstration evidence.

Configuration Reference

Item	Value
Target table	Employee Test / u_employee_test
Import Set table	Employee Import / u_employee_import
Transform Map	Sample Spreadsheet Import
Coalesce field	Employee ID
Dashboard	Employee Analytics Dashboards
Source document	Import Data using Transform Maps(3).docx
Team roles	Not supplied; all five members are listed on the cover.

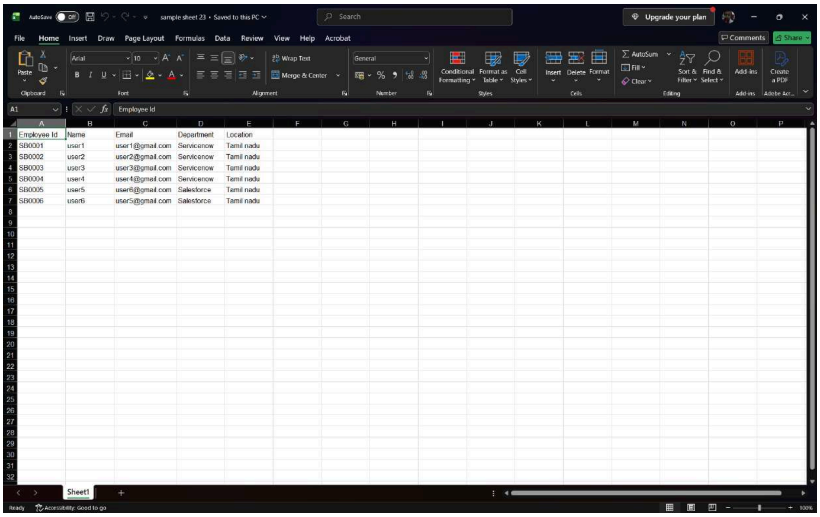


Figure 15. Source spreadsheet used to test changed and new employee records.